

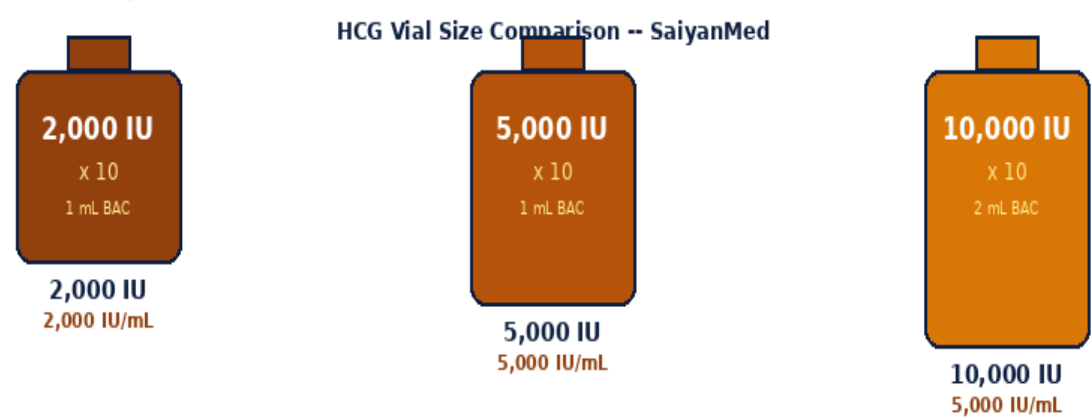
HUMAN CHORIONIC GONADOTROPIN

HCG / hCG | 2,000 IU -- 5,000 IU -- 10,000 IU | 3 VIAL SIZES

Gonadotropin Hormone -- Testosterone Support, PCT & Fertility Research
Purity: 99.8% | Endotoxin-Free | Research Use Only

Human Chorionic Gonadotropin (HCG) is a glycoprotein hormone that mimics Luteinizing Hormone (LH), stimulating Leydig cells in the testes to produce testosterone and maintain testicular size and function. It is widely researched in the context of testosterone replacement therapy (TRT) co-administration, post-cycle therapy (PCT), hypogonadism treatment, and fertility research. Available in three vial sizes: 2,000 IU, 5,000 IU, and 10,000 IU.

Vial Size Comparison



Vial	BAC Water	IU/mL	Doses (500IU ea)	Doses (1000IU ea)	Per Kit (x10)
2,000 IU x10	1 mL	2,000	4	2	40 doses (500IU)
5,000 IU x10	1 mL	5,000	10	5	100 doses (500IU)
10,000 IU x10	2 mL	5,000	20	10	200 doses (500IU)

MECHANISM OF ACTION & RESEARCH BENEFITS

HCG binds to LH receptors (LHCGR) on Leydig cells in the testes, directly stimulating steroidogenesis — the conversion of cholesterol to testosterone. Because HCG acts identically to LH at the receptor level, it effectively replaces the LH signal suppressed by exogenous androgens or other hormonal interventions. This makes it uniquely valuable in maintaining testicular function during TRT and restoring the hypothalamic-pituitary-gonadal (HPG) axis after anabolic suppression.

Pathway	Mechanism	Research Application
LH Receptor Activation	HCG binds LHCGR on testicular Leydig cells; activates adenylate cyclase via G-protein coupling; raises intracellular cAMP; activates protein kinase A; stimulates StAR protein and CYP11A1 enzyme for cholesterol conversion	Testosterone production restoration; maintains intratesticular testosterone during TRT
Testicular Maintenance	LH stimulation maintains Leydig cell population and testicular volume; absence of LH signal (e.g. during exogenous androgen use) causes Leydig cell atrophy and testicular shrinkage within weeks	Prevents testicular atrophy during TRT; preserves fertility and organ integrity
Spermatogenesis Support	Intratesticular testosterone (ITT) is required for normal spermatogenesis at ~80x higher concentrations than serum; HCG maintains ITT when LH is suppressed	Fertility preservation during hormonal therapy; male fertility research
HPG Axis Restoration (PCT)	After anabolic suppression, HCG rapidly restores Leydig cell sensitivity and testicular function; primes the axis for recovery before SERM therapy; prevents prolonged hypogonadism post-cycle	Post-cycle therapy (PCT); hypogonadism recovery research
Ovulation Induction (Female)	In females, HCG triggers final oocyte maturation and ovulation by mimicking LH surge; supports luteal phase via corpus luteum stimulation	Female fertility research; IVF/ART protocols

Key Research Benefits

Benefit	Evidence & Research Context
Testosterone Restoration	HCG directly stimulates Leydig cell testosterone production via LH receptor activation, restoring endogenous testosterone rapidly even after prolonged suppression.
Fertility Preservation	Maintains intratesticular testosterone and spermatogenesis during androgen therapy, preserving fertility in research subjects on long-term TRT.
PCT Efficacy	HCG is the most reliable agent for rapidly restoring testicular function post-cycle, reducing recovery time from months to weeks when used correctly.
Testicular Volume	Prevents and partially reverses testicular atrophy associated with HPG axis suppression — an important consideration in long-term hormonal research.

RECONSTITUTION PROTOCOL

Always use Bacteriostatic Water (BAC Water). HCG is a fragile glycoprotein — handle gently. Add BAC Water against the glass wall and never shake. The 2,000 IU and 5,000 IU vials use 1 mL BAC Water. The 10,000 IU vial uses 2 mL BAC Water. Both yield 5,000 IU/mL for the larger vial and their respective concentrations for the smaller vials.



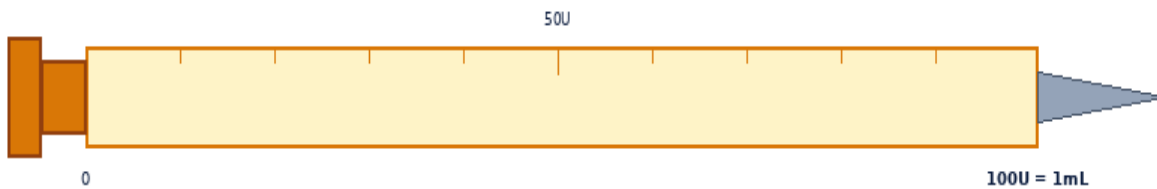
Swirl gently -- NEVER shake. Solution must be clear before use.

Step	Action	Detail
1	Bring to room temperature	Remove HCG vial and BAC Water from fridge. Allow both to reach room temperature (~20 deg C). HCG is a fragile glycoprotein -- temperature shock during reconstitution can reduce potency.
2	Clean stoppers	Wipe rubber stoppers of BOTH vials with a fresh alcohol swab. Air-dry 10-15 seconds before proceeding.
3	Draw BAC Water	Draw 1 mL for 2,000 IU or 5,000 IU vials. Draw 2 mL for 10,000 IU vials. Remove all air bubbles before proceeding.
4	Inject BAC Water into HCG vial	Direct stream against the GLASS WALL -- never directly onto the lyophilised powder. HCG protein will dissolve as BAC Water flows down the glass wall gently.
5	Dissolve gently	Roll vial slowly between palms. HCG dissolves within 30-60 seconds. NEVER shake, vortex, or agitate. Solution should be clear and colourless.
6	Inspect and label	Discard if cloudy or particulate. Write reconstitution date. Store at 2-8 deg C. Use within 28 days. Never freeze.

Reconstitution Quick Reference

Vial	BAC Water	IU/mL	250IU	500IU	1,000IU	2,500IU	5,000IU
2,000 IU	1 mL	2,000	0.125mL 12.5U	0.25mL 25U	0.50mL 50U	1.25mL --	N/A
5,000 IU	1 mL	5,000	0.05mL 5U	0.10mL 10U	0.20mL 20U	0.50mL 50U	1.00mL 100U
10,000 IU	2 mL	5,000	0.05mL 5U	0.10mL 10U	0.20mL 20U	0.50mL 50U	1.00mL 100U

DOSE CALCULATION & INJECTION TECHNIQUE



Insulin Syringe -- 1mL / 100U | Use 29-31G needle for SubQ

Formula: Volume (mL) = Desired Dose (IU) / Concentration (IU/mL) | **Units:** Volume (mL) x 100 = Units on insulin syringe

HCG Dose Range by Research Protocol



Vial: 2,000 IU x 10 -- Add 1 mL BAC Water → Concentration: 2,000 IU/mL

Dose (IU)	500 IU	1,000 IU	1,500 IU	2,000 IU	2,500 IU	5,000 IU
Volume (mL)	0.2500	0.5000	0.7500	1.0000	1.2500	2.5000
Syringe Units	25.0U	50.0U	75.0U	100.0U	125.0U	250.0U

Vial: 5,000 IU x 10 -- Add 1 mL BAC Water → Concentration: 5,000 IU/mL

Dose (IU)	500 IU	1,000 IU	1,500 IU	2,000 IU	2,500 IU	5,000 IU
Volume (mL)	0.1000	0.2000	0.3000	0.4000	0.5000	1.0000
Syringe Units	10.0U	20.0U	30.0U	40.0U	50.0U	100.0U

Vial: 10,000 IU x 10 -- Add 2 mL BAC Water → Concentration: 5,000 IU/mL

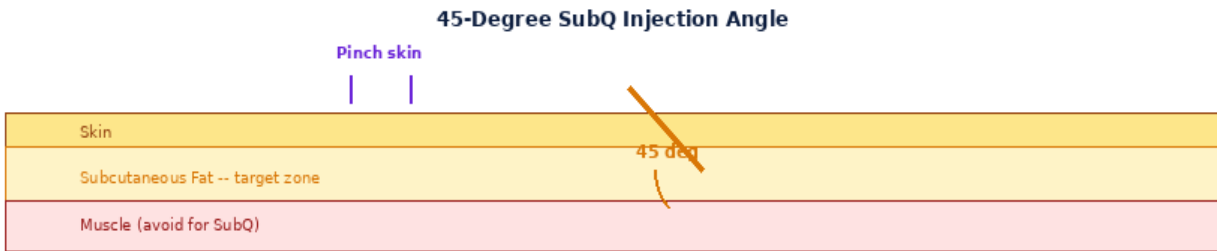
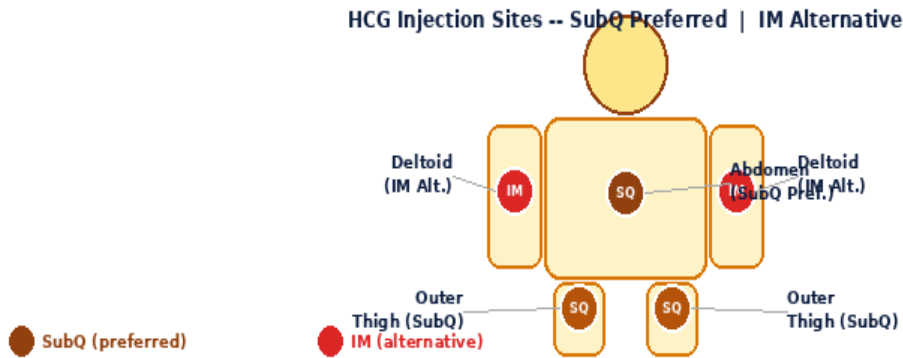
Dose (IU)	500 IU	1,000 IU	1,500 IU	2,000 IU	2,500 IU	5,000 IU
Volume (mL)	0.1000	0.2000	0.3000	0.4000	0.5000	1.0000
Syringe Units	10.0U	20.0U	30.0U	40.0U	50.0U	100.0U

Step-by-Step Injection Procedure

Step	Action	Detail
1	Wash hands	Wash 20 seconds with soap and warm water. Dry completely.
2	Select needle	SubQ: 29-31G x 5/16 inch insulin syringe (1 mL). IM: 23-25G x 1 to 1.5 inch.
3	Draw dose	Draw exact volume per dose table. Expel air bubbles.

4	Clean site	Wipe site with alcohol swab. Allow 10-15 seconds to dry.
5	Pinch and inject (SubQ)	Pinch 2-3 cm skin. Insert at 45 degrees. Release. Inject slowly.
6	Withdraw and press	Remove at same angle. Press with dry swab 5-10 sec. Do NOT rub.
7	Rotate and dispose	Record site used. Rotate systematically. Needle into sharps container.

INJECTION SITES, PROTOCOLS & SIDE EFFECTS



Alternative: Every Other Day (EOD). Dose depends on protocol -- see table.

Research Dosing Protocols

Protocol	Dose	Frequency	Duration	Key Notes
TRT Co-admin (Maintenance)	250-500 IU	EOD or 3x/week	Ongoing with TRT	Prevents atrophy; maintains ITT and fertility during TRT
PCT Phase 1 (Standard)	500-1,000 IU	3x/week	2-3 weeks	Start immediately post-cycle. Follow with SERM therapy (Nolvadex/Clomid).
PCT Phase 1 (Intensive)	1,000-2,500 IU	3x/week	2-3 weeks	After prolonged/heavy suppression. Monitor estrogen -- HCG aromatises.
Fertility (Male)	2,500-5,000 IU	2-3x/week	3-6 months	For spermatogenesis induction. Often combined with FSH analogue.
Hypogonadism (Research)	1,500-2,000 IU	3x/week	Until recovery	Secondary hypogonadism where LH pulsatility is impaired.

Off-cycle Break	0 IU	Pause	As needed	4-6 week minimum breaks between HCG courses to preserve receptor sensitivity.
-----------------	------	-------	-----------	---

Side Effects

Side Effect	Likelihood	Management
Estrogen elevation (aromatisation)	Common	HCG stimulates testosterone which aromatises to estradiol. Monitor E2. Use aromatase inhibitor if gynecomastia or high E2 symptoms.
Testicular ache / heaviness	Common	Normal at onset -- Leydig cells activating. Usually resolves within 1-2 weeks. Reduce dose if persistent.
Water retention / bloating	Dose-related	Oestrogen-mediated. Monitor dose. Reduce if significant.
Desensitisation (LH receptor)	High dose / chronic	Prolonged high-dose HCG causes LH receptor downregulation. Limit continuous use to 4-6 weeks. Take mandatory breaks.
Mood changes / irritability	Occasional	Related to testosterone/estrogen fluctuations. Monitor and manage E2.
Injection site reaction	Occasional	Rotate sites. Use fresh 29-31G needle. Warm vial to room temp first.
Ovarian hyperstimulation (female)	Dose-dependent	Risk of OHSS at high doses in females. Requires medical supervision in fertility protocols. Strict dose control essential.

STORAGE, GOLDEN RULES & SAFETY

STOP & SEEK MEDICAL ADVICE: Severe gynecomastia or breast tenderness not resolving with aromatase inhibitor

STOP & SEEK MEDICAL ADVICE: Signs of OHSS in females: severe abdominal pain, bloating, nausea, vomiting

STOP & SEEK MEDICAL ADVICE: Severe allergic reaction: hives, facial swelling, difficulty breathing

STOP & SEEK MEDICAL ADVICE: Signs of injection site infection: spreading redness, warmth, pus, fever

STOP & SEEK MEDICAL ADVICE: Prolonged suppression of natural testosterone production after HCG cessation

Absolute Contraindications: Androgen-sensitive tumours (prostate, testicular), known allergy to HCG or any component, precocious puberty, ovarian cysts (not related to PCOS) in female protocols, pregnancy contraindicated for exogenous HCG in certain contexts. Not for use in individuals under 18.

Storage Guidelines

State	Temperature	Duration	Notes
Unreconstituted powder	2-8 deg C	Per expiry	Keep upright. Away from light. Do not freeze.
Reconstituted solution	2-8 deg C	28 days	Label with reconstitution date. Inspect before each use.
BAC Water (opened)	2-8 deg C	28 days	Label with opening date.
Transport	Below 25 deg C	Max 72 hr	Insulated bag with ice packs. Never freeze.

Golden Rules -- HCG

1. NEVER shake the HCG vial -- it is a fragile glycoprotein; agitation destroys potency.
2. Direct BAC Water against the glass wall, never onto the powder.
3. Use 1 mL BAC Water for 2,000 IU and 5,000 IU vials; 2 mL for 10,000 IU vials.
4. HCG stimulates aromatisation -- monitor estradiol and use an AI if E2 rises significantly.
5. Do NOT use HCG continuously for more than 4-6 weeks without a break -- LH receptor desensitisation.
6. Rotate SubQ injection sites through all 8 abdomen points systematically.
7. For PCT: begin HCG immediately at end of cycle, before SERM therapy.
8. Store reconstituted HCG at 2-8 deg C. Never freeze. Discard after 28 days.
9. Contraindicated in androgen-sensitive cancers (prostate/testicular). Screen before use.
10. In female fertility protocols, strict dose control is essential to prevent OHSS.

DISCLAIMER: This document is produced by SaiyanMed for informational and research reference purposes only. Human Chorionic Gonadotropin (HCG) is a prescription substance in many jurisdictions. This guide does not constitute medical advice. HCG must not be used therapeutically without qualified medical supervision. SaiyanMed assumes no liability for misuse of this product or information herein.

SaiyanMed | saiyanmed.com | Research Use Only | For Educational Purposes